Project Proposal

AI-Assisted Society Prioritization System for Solar Energy Distribution & Outreach

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1. Problem Definition Statement

The Client operates solar farms that generate surplus electricity. Nearby housing societies in Maharashtra currently buy electricity from standard grid/DISCOM providers at a materially higher per-unit rate. The Client can supply the same electricity directly at a lower rate while still operating profitably - but today, deciding which societies to approach, and in what order, is done manually and is slow and inconsistent.

The problem this project addresses is:

> ***"Given a solar farm's available capacity and data on societies across Maharashtra (their electricity consumption and contact details), which societies should be approached first, and in what combination, to make the best use of that capacity?"***

Product Overview

1.1 Users

- **Client (Solar Energy Provider / Sales Team):** the primary, direct user of the system. Uses the Prioritized Outreach List, the Optimal Allocation List, and the dashboard to plan and sequence pitches to societies.
- **Housing Society Secretary (Decision-Maker):** the indirect user / target of outreach. Not a direct user of the software, but their society's consumption and contact data drive the ranking, and they are the eventual recipient of the Client's pitch.

1.2 Functionality

- Automated collection of society-level electricity consumption data across Maharashtra from the Client's identified data sources.
- Automated collection of society secretary / decision-makers contact details from a separate identified source.
- Cleaning, deduplication, and consolidation of the above into a single structured dataset.
- A configurable, rule-based scoring and ranking engine, built with the Client's real prioritization criteria.
- A constrained optimization (bin-packing style) engine that matches combinations of societies to a farm's available capacity.

- Scheduled, automatic refresh of outputs as underlying data changes.
- Presentation of results through a non-technical-friendly interface: spreadsheet/CSV output plus a lightweight dashboard.

1.3 Features

- Prioritized Outreach List - ranked societies answering “who should we talk to first?”
- Optimal Allocation List - best combination of societies to fully utilize a given farm's capacity, answering “which societies should we target together?”
- Lightweight Streamlit dashboard for browsing and filtering results.
- Configurable rule engine - ranking criteria are not hard-coded, since they are expected to evolve.
- Scheduled (cron-style) refresh job with a configurable interval.
- (Phase 2) ML-based ranking model trained on real pitch outcomes, with the Phase 1 rule-based logic retained as a baseline/fallback.

Proposed Solution

The project is delivered in two phases. Phase 1 builds a data pipeline and a rule-based/optimization-based prioritization system; Phase 2 layers machine learning on top of it once real outcome data is available. The first milestone in Phase 1 is a short discovery spike to confirm the availability, structure, and reliability of the Client's data sources before the main build begins.

- **Data ingestion layer:** connectors/scripts for each confirmed consumption and contact-detail source (download, API, or scraping, depending on what the Client provides).
- **Processing layer:** cleaning, deduplication, and consolidation into one structured dataset covering societies across Maharashtra.
- **Ranking/optimization layer:** a configurable rule engine producing both output lists in Phase 1, extended with a trained scoring model as pitch feedback accumulates in Phase 2.
- **Output layer:** CSV/spreadsheet exports plus a Streamlit dashboard for browsing and filtering.
- **Scheduling layer:** a configurable, time-based refresh job so outputs stay current as underlying data changes.

Default ranking rules for Phase 1 (e.g. matching consumption against available capacity, proximity to the farm, and estimated cost savings versus the current provider) will be used as a starting point and refined once the Client confirms specific criteria.

Expected Outcome & Future Plan

3.1 Expected Outcome

- A reliable, refreshable Prioritized Outreach List and Optimal Allocation List that the Client can act on directly.
- A simple, non-technical dashboard that makes the outputs usable without engineering support.
- A clean, extensible foundation so later capability (ML scoring, and beyond) can be added without rebuilding the system.

- (Phase 2) Improved prioritization accuracy over the Phase 1 rule-based approach, driven by real pitch outcomes, with the rule-based logic kept as a baseline for comparison.

3.2 Future Plan

This proposal covers Phase 1 (data pipeline and rule-based/optimization-based prioritization) and Phase 2 (ML-based scoring refinement). Beyond Phase 2, natural-language reasoning, RAG/NL2SQL, and automated outreach are promising directions the project could grow into, but are intentionally out of scope for now - they will be discussed and scoped separately once Phase 2 has been delivered and proven out.

Challenges

- **Data source confirmation pending:** the exact access method (download, API, or scraping) for consumption and contact-detail sources is not yet confirmed by the Client; this is the first dependency to resolve and may affect the technical approach and timeline.
- **Ranking criteria pending:** the Client has not yet specified detailed prioritization rules; Phase 1 will start with reasonable defaults and refine them once confirmed.
- **Data reliability:** the quality, completeness, and update-frequency of third-party data sources is unknown until inspected, and will be assessed during the discovery spike.
- **Data privacy & confidentiality:** the project involves personal data (society secretaries' names and contact details), which must be handled per India's Digital Personal Data Protection Act, 2023 (DPDP Act) - collected from lawful sources, stored securely, restricted to students actively working on the relevant module, and used strictly for this project's stated purpose.

References (if any)

- India's Digital Personal Data Protection Act, 2023 (DPDP Act).
- Client-provided data sources for society-level electricity consumption and secretary/contact details across Maharashtra (links and access details pending).